FULL STACK APPLICATION DEVELOPMENT

#SKILL - 2 Hibernate CRUD Operations Prerequisites:

- Basic knowledge of Java
- Understanding of classes, objects, and OOP
- Familiarity with SQL basics

A retail inventory system needs to store product details such as productId, name, description, price, and quantity. The admin should be able to add new products, retrieve product information, update price or quantity, and delete discontinued products.

Your task is to implement complete CRUD operations using Hibernate with proper entity mapping using JPA annotations.

1. Create a Product entity with fields: id (primary key), name, description, price, quantity.
2. Configure Hibernate and map the Product entity to a table using JPA annotations.
3. Insert multiple Product records into the database.
4. Retrieve a product using its id.
5. Update the price or quantity of any selected product.
6. Delete a product record by id if it is discontinued.
7. Note:

a. Implement Hibernate ID generation strategies (AUTO, IDENTITY, SEQUENCE) inside the entity to observe how primary key values di er.

b. The ID can be manual or auto generated, based on your chosen strategy.

8. Push the Hibernate project into a single GitHub repository.

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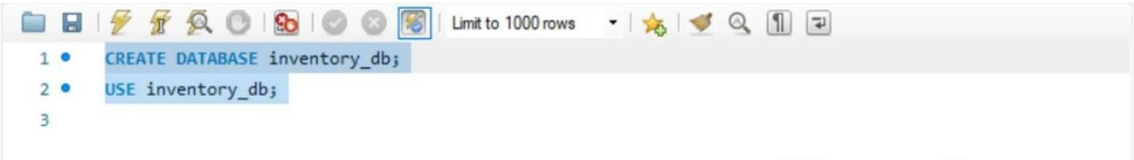
Create Database (IMPORTANT)

1. Open MySQL Command Line Client (or MySQL Workbench)
2. Login using password
3. Run this:

```
CREATE DATABASE inventory_db;

USE inventory_db;
```



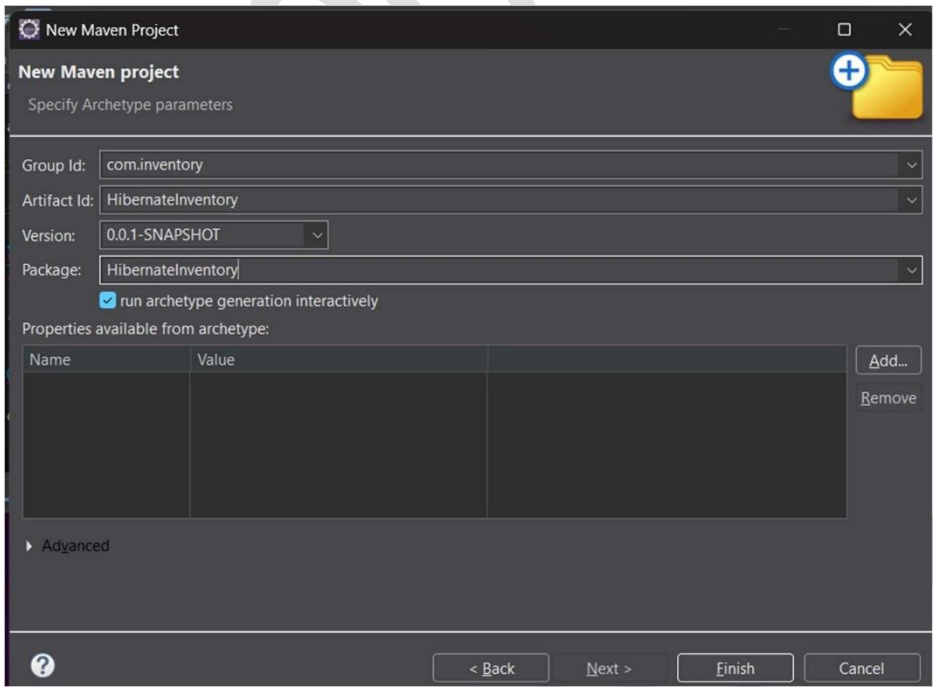
Create Maven Project in Eclipse

1. In Eclipse menu:

File → New → Maven Project

2. Click Next
3. 膝 Tick: Create a simple project (skip archetype selection)
4. Click Next

Field	Value
Group Id	com.inventory
Artifact Id	HibernateInventory
Packaging	jar
Name	HibernateInventory



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Click Finish

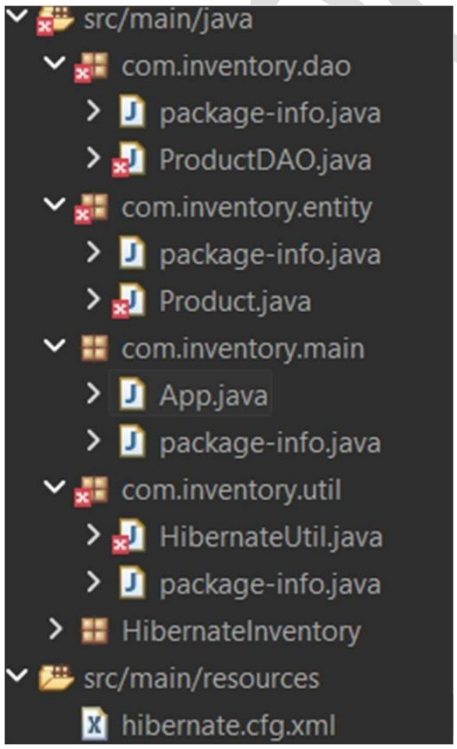
Add Maven Dependencies

1. Open pom.xml

2.paste this inside <dependencies>

```
        <dependency>
  <groupId>junit</groupId>
  <artifactId>junit</artifactId>
  <version>3.8.1</version>
</dependency>
```

Create Packages



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Product.java

package com.inventory.entity; import jakarta.persistence.*; @Entity

@Table(name = "products") public class Product {

@Id

@GeneratedValue(strategy = GenerationType.AUTO)

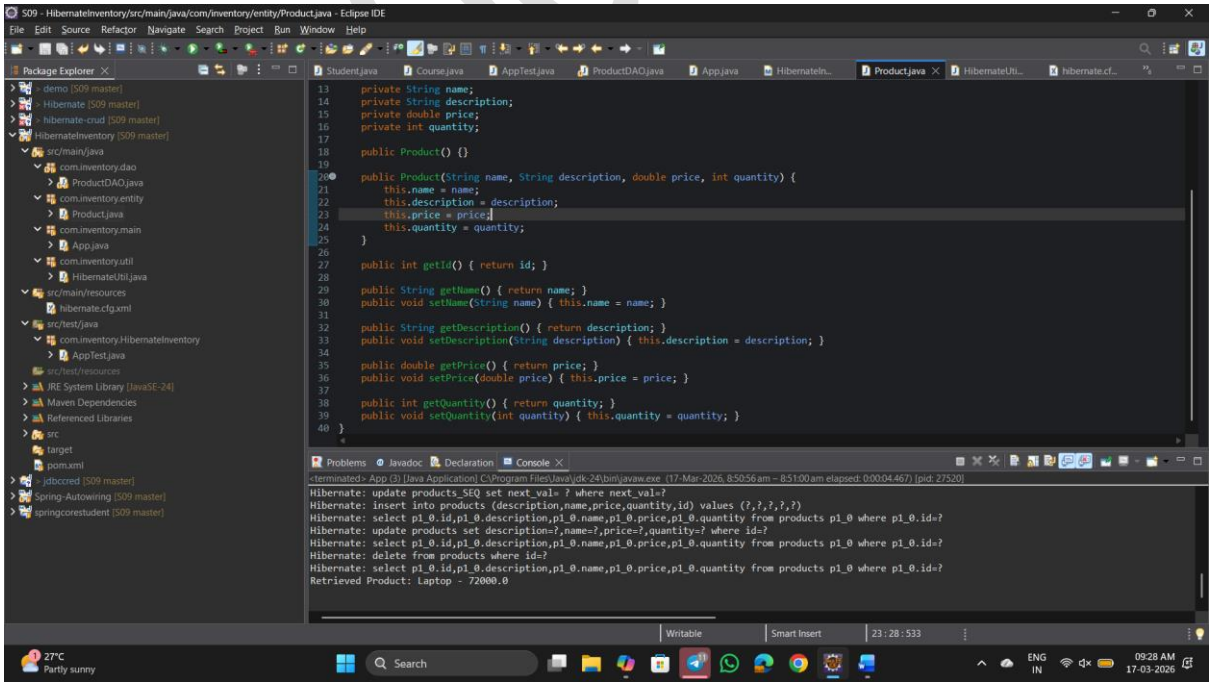
private int id; private String name; private String description; private double price; private int quantity;

public Product() {} public Product(String name, String description, double price, int quantity) { this.name = name; this.description = description; this.price = price; this.quantity = quantity;

}

public int getId() { return id; } public String getName() { return name; } public void setName(String name) { this.name = name; } public String getDescription() { return description; } public void setDescription(String description) { this.description = description; } public double getPrice() { return price; } public void setPrice(double price) { this.price = price; } public int getQuantity() { return quantity; } public void setQuantity(int quantity) { this.quantity = quantity; }

}



HibernateUtil.java

```
package com.inventory.util; import org.hibernate.SessionFactory; import
org.hibernate.cfg.Configuration; public class HibernateUtil {    private static SessionFactory
factory;  static {        factory = new Configuration().configure().buildSessionFactory();

    }

public static SessionFactory getSessionFactory() {        return factory;

import org.hibernate.Session; import org.hibernate.Transaction; import
com.inventory.entity.Product;

import com.inventory.util.HibernateUtil; public class ProductDAO {

    public void saveProduct(Product product) {

        Session session = HibernateUtil.getSessionFactory().openSession();        Transaction
tx = session.beginTransaction();        session.save(product);        tx.commit();
session.close();

    }

    public Product getProduct(int id) {

        Session session = HibernateUtil.getSessionFactory().openSession();        Product
product = session.get(Product.class, id);        session.close();        return product;

    }

    public void updateProduct(int id, double price, int quantity) {

        Session session = HibernateUtil.getSessionFactory().openSession();

        Transaction tx = session.beginTransaction();        Product product =
session.get(Product.class, id);        product.setPrice(price);
product.setQuantity(quantity);        session.update(product);        tx.commit();
session.close();

    }

    public void deleteProduct(int id) {

        Session session = HibernateUtil.getSessionFactory().openSession();

        Transaction tx = session.beginTransaction();

 Product product = session.get(Product.class, id);        session.delete(product);
tx.commit();        session.close();

    }


import com.inventory.dao.ProductDAO; import com.inventory.entity.Product; public class App
{

    public static void main(String[] args) {

        ProductDAO dao = new ProductDAO();
```

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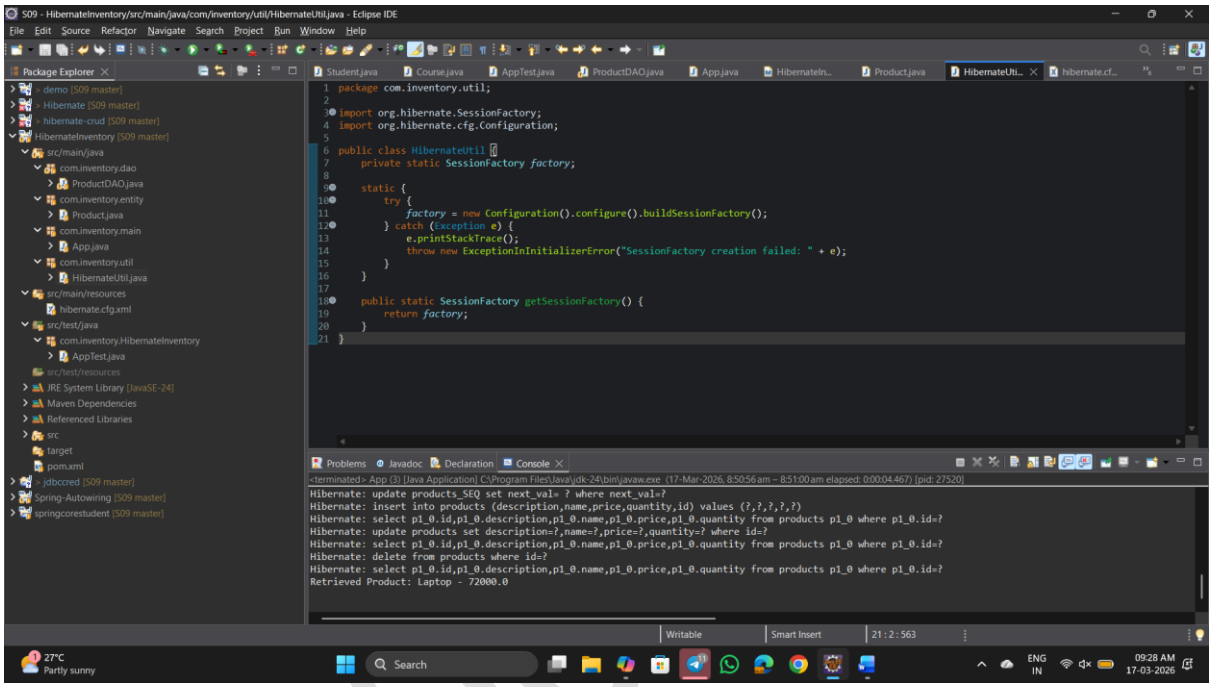
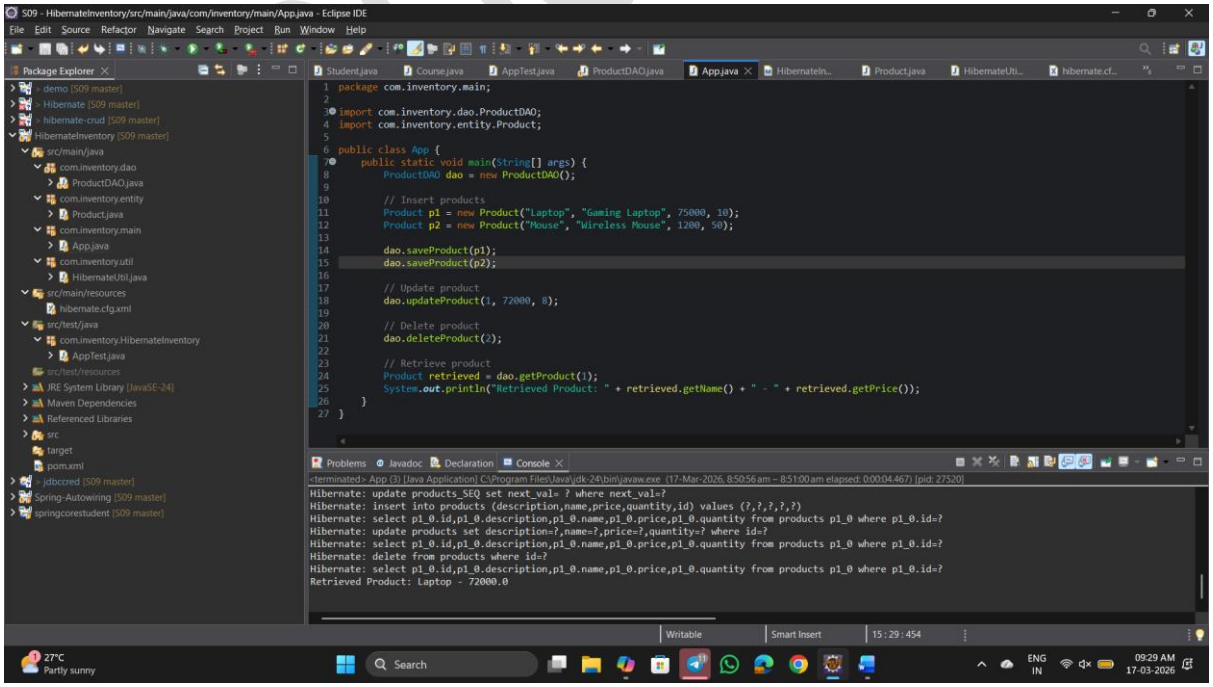
Product p1 = new Product("Laptop", "Gaming Laptop", 75000, 10); Product p2 = new Product("Mouse", "Wireless Mouse", 1200, 50); dao.saveProduct(p1); dao.saveProduct(p2);

dao.updateProduct(1, 72000, 8); dao.deleteProduct(2);

}

}





Create Hibernate Configuration File

1. Right-click src/main/resources
2. New → File
3. File name: hibernate.cfg.xml
4. Click Finish
5. Paste this:

```
<!DOCTYPE hibernate-configuration PUBLIC
        "-//Hibernate/Hibernate Configuration DTD 3.0//EN"
        "http://hibernate.sourceforge.net/hibernate-configuration-3.0.dtd">
<hibernate-configuration>
    <session-factory>
<property
name="hibernate.connection.driver_class">com.mysql.cj.jdbc.Driver</property>
        <property name="hibernate.connection.url">
jdbc:mysql://localhost:3306/inventory_db
        </property>
        <property name="hibernate.connection.username">root</property>
        <property name="hibernate.connection.password">YOUR_PASSWORD</property>
 <property name="hibernate.dialect">org.hibernate.dialect.MySQLDialect</property>
        <property name="hibernate.hbm2ddl.auto">update</property>
        <property name="hibernate.show_sql">true</property>
 <mapping class="com.inventory.entity.Product"/>
    </session-factory>
```

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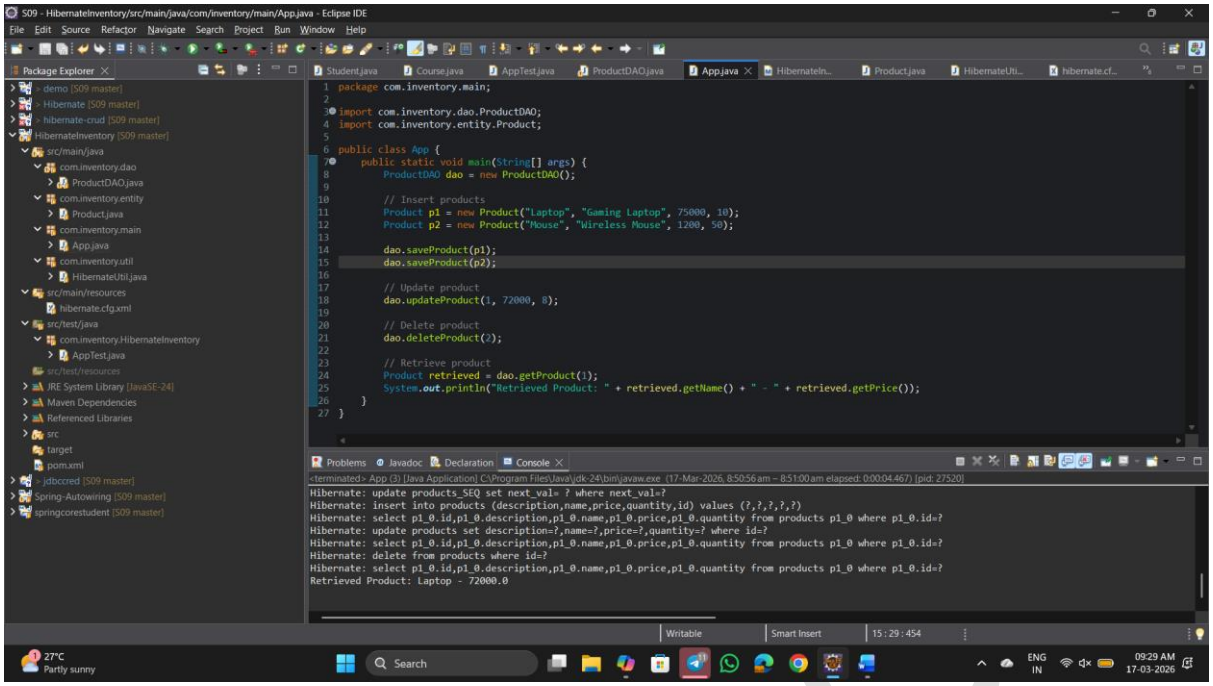
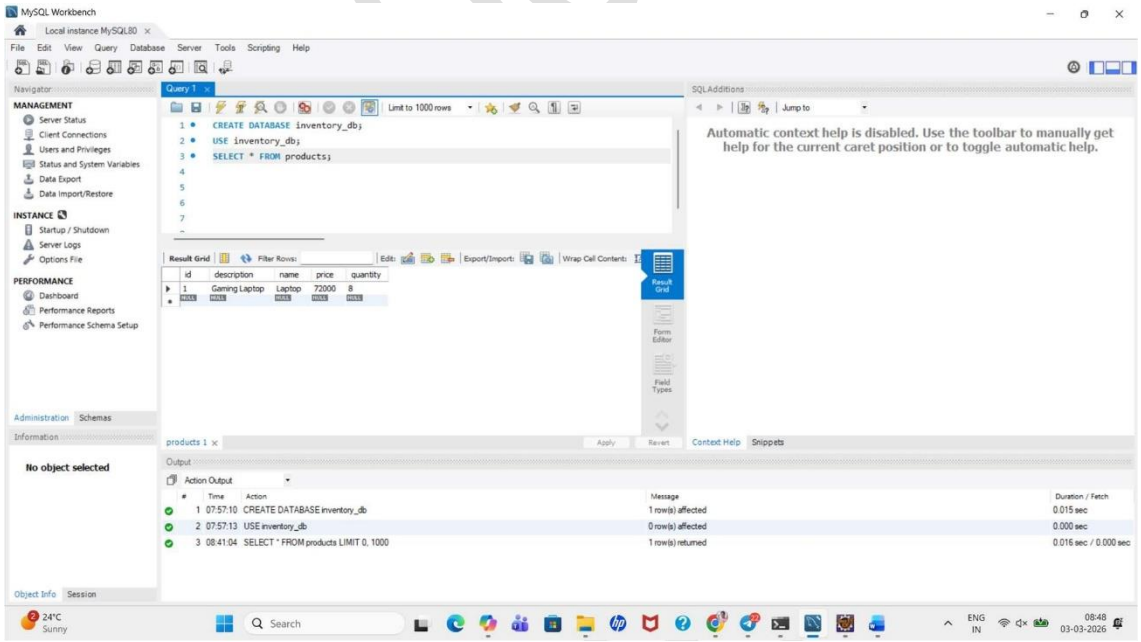


Run the Project

1. Right-click App.java
2. Run As → Java Application

Output:



Git url : https://github.com/Eswar2400032797/git-skill2.git